

ELCIA

Hydra Sense

**TRL-8 Validated Cold Chain
Monitoring**

Team Members-

**LIKITH
TARAL
VINAY**

**IIT
BANGALORE**

Ensuring Medical Integrity from Lab to Last-mile



When cooling fails, Lives are at risk

- Every year live saving vaccines, medicines and organs in India are wasted.
- The main reason? cooling breaks that no one notices.
- This “silent failure” means waste of medicine, putting patients at risk.

Hydra-Sense for Medicine Safety

- We created an affordable device using readily available sensors.
- It monitors Temperature and Humidity all the time.
- If something feels off, alerts will be sent and data can be checked anywhere through cloud.

A Self-Reliant Future for Healthcare

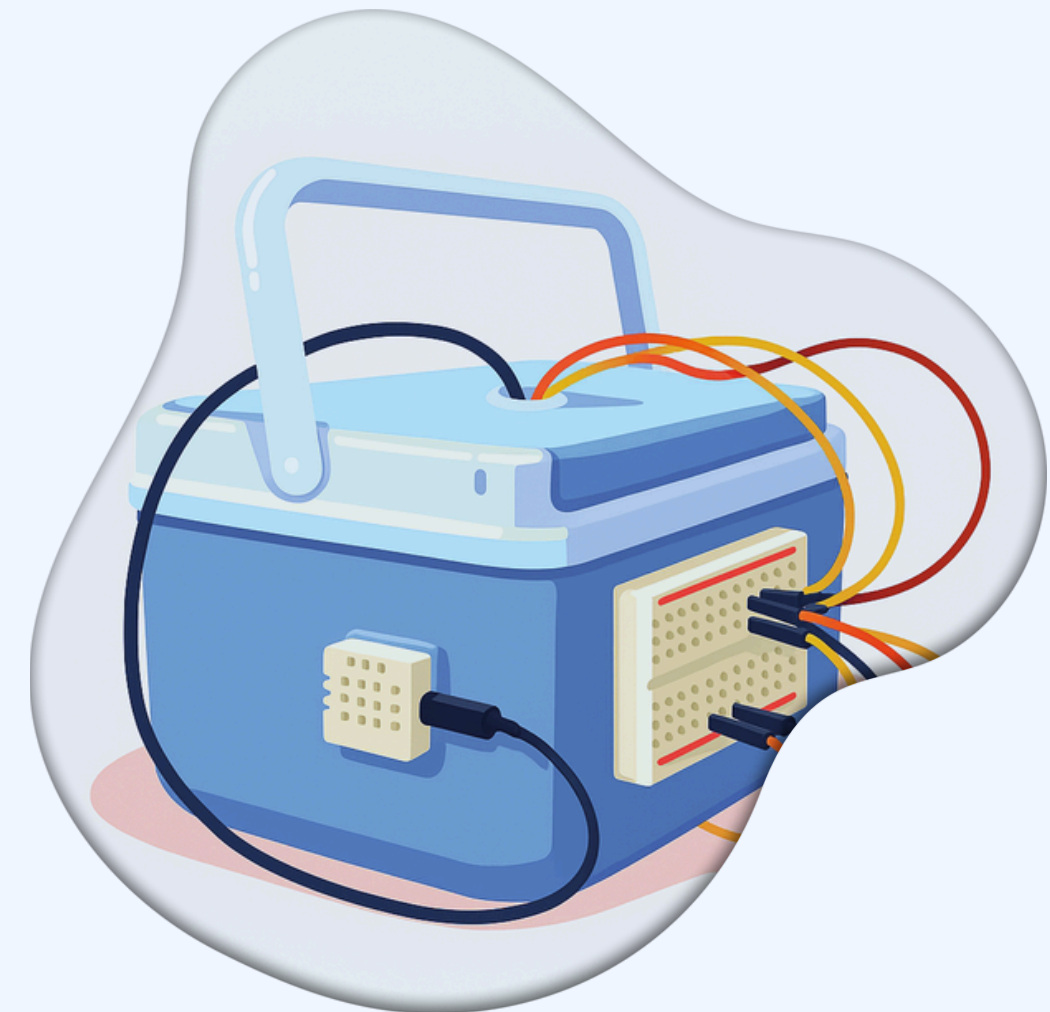
- Made in India Sensor Integration.
- DHT22 – SHT31-D
- DS18B20 – PT100 RTD
- B,E680 – SEN54

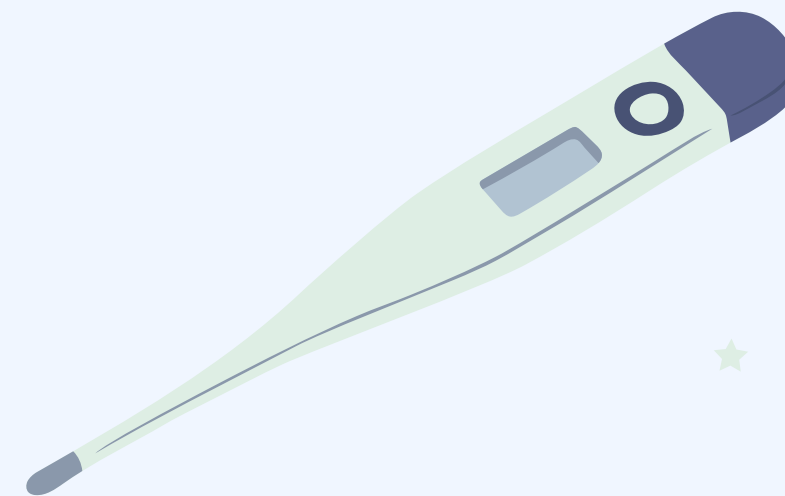
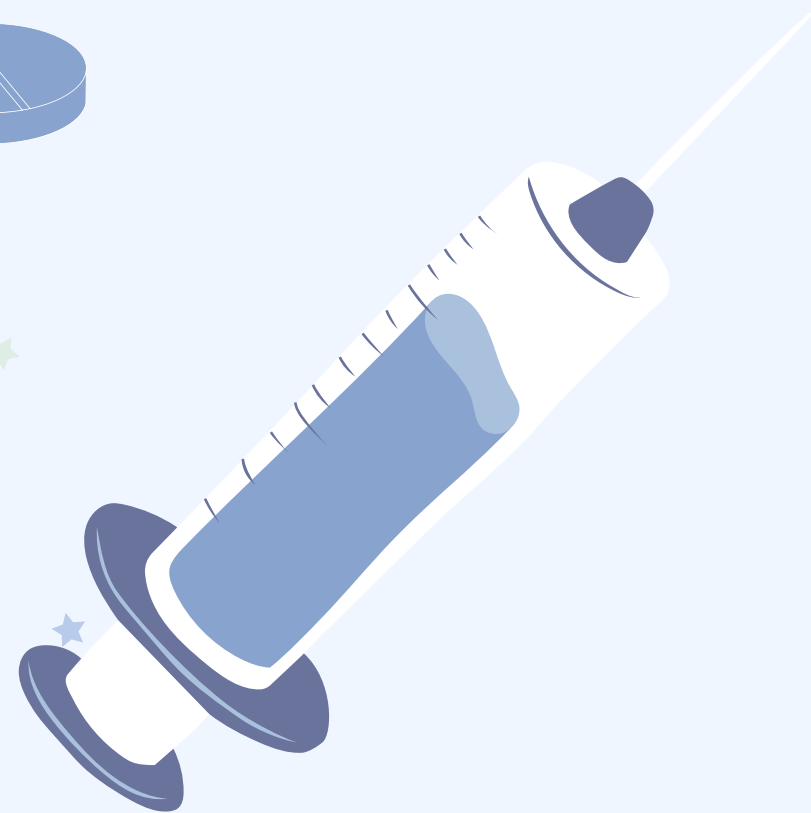
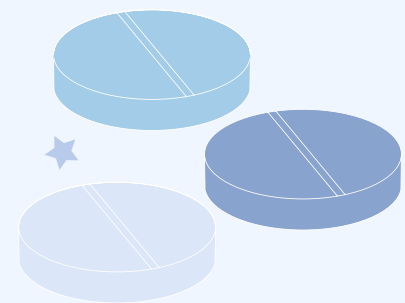
From Prototype to Smart Network

- Multi-location sensor integration.
- Making B2B and B2C partnerships.
- Use of Machine Learning for early failure.

Tested like it's Real-Life

- 16 Hours Drift test.
- Calibrated for high and low humidity.
- Cold to Hot transition and repeated power on/off cycles.





Thank You



Kanyarashi